

EXPERIMENT 20:

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C:\Users\trish\OneDrive\Documents\exp20.R - R Editor
# 2. Automatically generate a 3rd feature (Insulin^2 or an index) so Multiple F
diabetes$Insulin_Sq <- (diabetes$Insulin)^2

# Select subset
input <- diabetes[, c("Age", "Insulin", "Insulin_Sq")]

# 3. Fit Multiple Linear Regression (Predicting Age from Insulin and Insulin_Sq)
model <- lm(Age ~ Insulin + Insulin_Sq, data = input)

# 4. Display model
print(model)

# 5. Extract coefficients
A <- coef(model)[1]          # Intercept
xInsulin <- coef(model)[2]    # Coefficient for Insulin
yInsulinSq <- coef(model)[3]  # Coefficient for Insulin_Sq

# 6. Print coefficients individually
print(A)
print(xInsulin)
print(yInsulinSq)

# 7. Compute prediction / linear sum
y <- A + xInsulin + yInsulinSq
print(y)
```

```
R Console
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Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 19.93 on 18 degrees of freedom
Multiple R-squared:  0.04957,    Adjusted R-squared:  -0.003228 
F-statistic: 0.9389 on 1 and 18 DF,  p-value: 0.3454

> source("exp20.R")

Call:
lm(formula = Age ~ Insulin + Insulin_Sq, data = input)

Coefficients:
(Intercept)      Insulin  Insulin_Sq 
  48.698043    0.019745   -0.002632 

(Intercept)
  48.69804
  Insulin
0.01974469
  Insulin_Sq
-0.002632047
(Intercept)
  48.71516
> |
```